

MSK-Bench: Benchmarking Full-Body Musculoskeletal Motor Control Across Tasks, Control Paradigms, and Physiological Metrics

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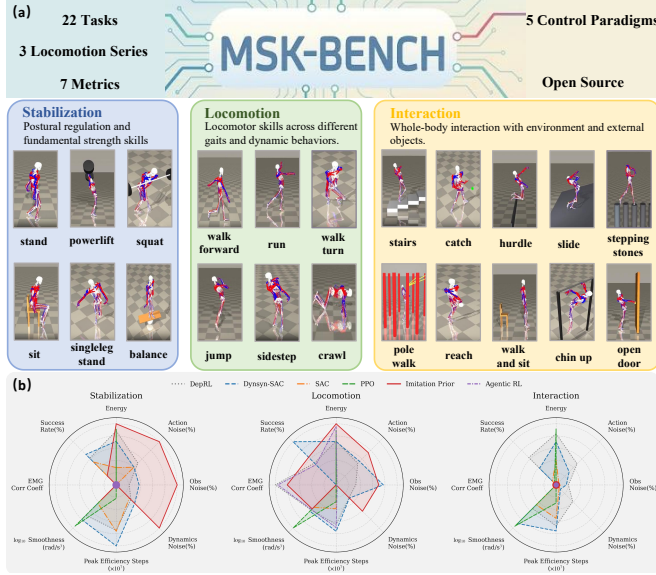


Fig. 1. **Overview of the Musculoskeletal Benchmark (MSK-Bench).** (a) **Benchmark Components:** An open-source suite featuring 22 tasks across stabilization, locomotion, and interaction, evaluated under 5 control paradigms and 7 metric families. (b) **Method Evaluation:** Radar charts compare control methods across the 3 task families using 8 reported axes derived from the benchmark metrics. Each value represents the average performance across all applicable tasks within the respective category. EMG-envelope similarity is evaluated only where matched human reference data are available, while cumulative reward is reported separately in Fig. 2.

Abstract—Musculoskeletal (MSK) humanoids provide a physiologically grounded embodiment for studying full-body motor control, but their high-dimensional muscle actuation, delayed activation dynamics, and redundant muscle-tendon structures make learning substantially harder than torque-driven humanoid control. Existing MSK benchmarks remain fragmented across gait, prosthetics, dexterous hands, or challenge-specific tracks, leaving full-body muscle-actuated control insufficiently evaluated under standardized tasks, methods, and metrics. We introduce MSK-Bench, a benchmark of 22 full-body motor-control tasks organized into three progressively challenging categories: postural stabilization, common locomotor behaviors, and contact-rich environmental interaction. Under unified task protocols and robustness perturbations, MSK-Bench evaluates 5 representative control paradigms, including reward-based RL, agentic reward tuning, latent-action RL, imitation-prior control, and residual adaptation over imitation priors. Beyond task success and reward, MSK-Bench further reports robustness analysis and physiology-oriented diagnostics, including activation cost, joint smoothness, and EMG-envelope similarity. Our empirical study shows that embodiment-aware exploration and structured action representations improve task coverage in high-dimensional muscle spaces, imitation priors enhance reference-compatible stabilization and locomotion but degrade under contact-rich terrain mismatch, and residual adaptation can recover successful behaviors when fixed references fail. We further find that improved task success does not

necessarily imply improved physiological agreement, highlighting the importance of evaluating task performance, robustness, and physiological behavior jointly. MSK-Bench provides a task-method-metric testbed for full-body muscle-actuated humanoid control. Project page and supplementary material: <https://anonymous.4open.science/r/MSK-Bench-FBB1/>; code and reproducibility package: <https://anonymous.4open.science/r/MSK-Bench-CF1B/>.

I. INTRODUCTION

Human motion emerges from the coordinated control of hundreds of muscles acting through a highly redundant and biomechanically constrained musculoskeletal system [1]. Unlike torque-driven humanoids, where actions directly specify joint torques, biological movement is generated through muscle excitation, activation-contraction dynamics and tendon transmission [2]. This physiological hierarchy introduces substantial control challenges including delayed actuation, muscle redundancy, and complex nonlinear dynamics. Consequently, musculoskeletal (MSK) models have become an increasingly important platform for studying physically grounded control [3].

Despite this progress, our understanding of how to learn effective control policies for large-scale musculoskeletal systems remains limited. Recent torque-driven humanoid benchmarks have substantially expanded task diversity for whole-body locomotion, loco-manipulation, and scene interaction, but they typically rely on rigid-body torque actuation (Tab. I). Conversely, existing MSK benchmarks provide physiologically grounded actuation, but they remain largely fragmented across specific domains, including gait optimization [4], prosthetics [5], athletic locomotion [6], and dexterous hand control [7], [8]. While these benchmarks have significantly advanced individual subfields, they are still limited in at least one of the dimensions needed for general-purpose full-body muscle-actuated control: broad whole-body task diversity, contact-rich terrain and object interaction, or standardized evaluation across multiple control paradigms. As a result, it remains unclear which control paradigms are most effective for full-body musculoskeletal agents and under what physical conditions they succeed or fail.

A second limitation of existing research lies in how musculoskeletal control is evaluated. Most existing benchmarks primarily focus on task-level metrics such as reward, success rate, or locomotion speed. However, unlike conventional humanoid control, musculoskeletal systems provide access to physiologically meaningful internal variables, including muscle activations, muscle synergies, and energy expenditure [9]. Multiple muscle patterns can often generate similar body

motions, implying that successful task completion alone is insufficient for evaluating musculoskeletal control. A policy may achieve the correct movement while relying on unrealistic muscle coordination strategies. Therefore, evaluating whether a policy behaves in a biomechanically plausible manner is equally important as evaluating whether it completes a task successfully [10].

To address these challenges, we propose **MSK-Bench**, a comprehensive full-body musculoskeletal benchmark designed to resolve current evaluation deficiencies: the lack of humanoid-level task diversity, the scarcity of rich physical interactions, and the absence of standardized physiological metrics (Tab. I provides a detailed comparison) [7], [10]–[16]. Specifically, MSK-Bench encompasses **22** tasks organized into three progressively challenging categories: **stabilization**, **locomotion**, and **complex interactions** (e.g., load transport and whole-body coordination), as illustrated in Fig. 1a. Moving beyond mere task success, we establish a physiology-aware evaluation protocol that assesses biomechanical fidelity through standardized metrics, including robustness, muscle activation cost, joint smoothness and EMG-envelope similarity. We evaluate **five** representative control paradigms under this unified protocol to provide a comprehensive multi-dimensional analysis of their control capabilities (Fig. 1b). This evaluation establishes a strong baseline, demonstrating that residual reinforcement learning effectively combines motor priors with task adaptation to offer a robust foundation for future research.

Through **MSK-Bench**, we identify three empirical findings about full-body muscle-actuated control. **First**, controller performance is strongly task- and metric-dependent: reward-based methods exhibit different advantages across task families and individual tasks, while focused studies reveal trade-offs in reward adaptation and action representation rather than a uniformly superior design. **Second**, motion-prior control is also task-dependent: the evaluated imitation controller achieves strong measured success on several stabilization and locomotion skills, but fails on the tested stair task even at zero dynamics perturbation; tracking errors characterize the resulting reference divergence without identifying its cause. **Third**, task-level success does not fully characterize muscle use. Physiology-aware evaluation reveals that success and EMG-envelope agreement can change differently across tasks, that agreement varies across individual muscles and control settings, and that recruitment profiles depend on task conditions and anatomical resolution. These findings motivate a diagnosis-to-validation workflow in which broad task evaluation identifies controller limitations, task-specific analysis characterizes them, and muscle-level diagnostics assess behavior beyond task completion.

II. RELATED WORK

Musculoskeletal Models and Control. Musculoskeletal (MSK) agents differ from torque-driven humanoids: muscle excitations are transformed by activation-contraction dynamics into muscle forces and then mapped to joint torques through tendon paths and moment arms, introducing redundancy, delay, nonlinear muscle-tendon dynamics, and

TABLE I
BENCHMARK COMPARISON. F/S/C/H/P: FULL BODY, SCENE/TERRAIN, CONTACT, HIERARCHY, PHYSIOLOGY. ✓/•/X: FULL/PARTIAL/NO SUPPORT IN CITED EVALUATIONS. K: CONTROL FAMILIES; EACH BENCHMARK LISTS ITS FOCUS.

Benchmark / focus	Actuation (dim.)	Scale	F/S/C/H/P	K
Torque-driven humanoids				
HumanoidBench [11] – Loco-manipulation	Torque 19/61	27 tasks	✓✓✓✓X	1
SkillBench/SkillBlender [12] – Skill blending	Torque 19	4 skills / 8 tasks	✓•✓✓X	1
Mimicking-Bench [13] – Scene interaction	Torque 19	6 tasks	✓✓✓X	1
Musculoskeletal benchmarks				
LocoMuJoCo [17] – Locomotion imitation	Mixed	12 envs / 27 tasks	✓X•X	2
MyoSuite [3] – Dexterity and agility	Muscle ≤ 80	204 tasks	XX•X	1
MyoDex [7] – Hand manipulation	Muscle 39	14 train / 34 eval.	XX•XX	1
MS-Human-700/MsGym [16] – MSK locomotion	Muscle 700	3 tasks	✓X•X	1
MyoChallenge 2024 [6] – Prosthetics	Muscle ≤ 80	2 tracks	•✓✓X	1
MyoChallenge 2025 [18] – Athletic control	Muscle ≤ 80	2 tasks / 4 tracks	•••X	1
MSK-Bench (ours) – Full-body terrain MSK	Muscle 416	22 tasks	✓✓✓✓✓	5

biomechanical constraints [2]. Classical platforms established workflows for anatomically grounded model construction, scaling, and motion-data analysis [2], while MyoSim and MyoSuite brought physiologically realistic MSK models into MuJoCo-style contact-rich RL settings [3], [19]. Recent work further advances MSK control through model-based muscle controllers, hierarchical planning, markerless-capture imitation, and physical musculoskeletal humanoids [8], while full-body systems have scaled toward large-scale motion imitation, richer anatomy, and whole-body behavior emulation [15], [16]. These advances motivate a systematic evaluation of full-body MSK control.

Benchmarks for Embodied Motor Control. Robot-learning benchmarks have standardized progress in continuous control, offline RL [20], and manipulation [21]. Recent humanoid benchmarks further move from isolated locomotion or manipulation toward whole-body interaction [11]–[14]. However, these benchmarks primarily target torque-driven rigid-body robots rather than physiological muscle actuation. For the MSK benchmark, early OpenSim-based challenges focused on gait, running, prosthetics, and neuromechanical locomotion control [4], [5], [17]. MyoSuite [3] and MyoChallenge [6] introduce contact-rich MSK tasks for physiologically realistic models, spanning hand manipulation, locomotion, and athletic intelligence tracks. Our benchmark organizes full-body MSK tasks into a graded spectrum from postural regulation to common locomotor behaviors and contact-rich environmental interaction. It evaluates multiple control paradigms under a task-method-metric matrix, where metrics go beyond task success to include EMG-envelope activation [10], [15] for assessing biomechanically plausible muscle coordination.

Learning Paradigms for Musculoskeletal Control. Reward-based RL provides a general route to motor control [22], [23], but the high dimensionality, redundancy, and delay of muscle actuation make direct exploration inefficient in MSK agents, motivating MSK-specific exploration, action-structure, and weak-prior methods [7], [24], [25]. Agentic reward generation further reduces manual reward engineering in robotics [26], yet its reliability under physiologically constrained MSK dynamics remains unclear. A complementary family relies on motion references and motor priors: refer-

ences can be reconstructed from human motion or generated by large motion models [27], while imitation and prior-based controllers have enabled both physics-based character control and muscle-actuated motion imitation [10], [15]. However, reference tracking alone may fail on contact-rich or out-of-distribution tasks, motivating residual adaptation that learns corrective actions on top of existing controllers or skill policies [28]. Our benchmark jointly evaluates reward-based RL, agentic reward tuning, latent-action RL, imitation-prior control, and residual adaptation under the same full-body MSK task suite.

III. MSK-BENCH

As shown in Fig. 1a, MSK-Bench evaluates three foundational functional regimes of gross full-body motor control: maintaining postural stability, generating self-propelled movement, and coordinating physical interaction with external terrain or objects [29]. These regimes progressively broaden the control demands from body stabilization to dynamic locomotion and environmental interaction. Accordingly, the 22 tasks are organized into three families: 1) **Stabilization** comprises six tasks—standing, single-leg standing, squatting, sitting, balancing, and powerlifting—that evaluate static or quasi-static postural control under body weight and external loading, emphasizing balance and trunk stabilization; 2) **Locomotion** comprises six tasks—walking, running, jumping, crawling, sidestepping, and turning—that evaluate self-propelled movement, requiring coordinated lower-limb recruitment, dynamic balance, and impact handling; 3) **Interaction** comprises ten tasks—stairs, hurdles, stepping stones, sliding terrain, reaching, catching, chin-up, door opening, pole walking, and walk-and-sit—that evaluate whole-body coordination under terrain variation and object contact, combining lower-body support, trunk stabilization, and task-dependent upper-limb or grip control. Together, these families assess whether muscle-actuated agents can balance, move, and interact with environment, while supporting physiological analysis across diverse movement regimes.

Experimental Setup. All 22 tasks use MuJoCo and the same 416-muscle elastic-tendon full-body model; the 700-muscle model is used only in the anatomy study. Tasks share a muscle-action interface and base proprioceptive observations, supplemented by task-dependent phase, terrain-height, or obstacle signals. Rewards combine task progress with regime-specific physical constraints.

Evaluation Protocol. Episode success requires $\mathcal{E}_{\text{task}}^{(i)} \wedge \mathcal{E}_{\text{alive}}^{(i)} \wedge \mathcal{E}_{\text{safe}}^{(i)}$; criteria and termination rules are fixed per task, with exact thresholds provided in the supplementary material. Within each task, the four full-suite baselines share observations, rewards, termination conditions, and evaluation. We evaluate seven complementary metric families across three dimensions: **task effectiveness and learning efficiency** (Success Rate, Cumulative Reward, and Peak-Efficiency Steps), **robustness** (Perturbation Robustness), and **physiological behavior** (Activation Cost, Joint Smoothness, and EMG-Envelope Similarity). 1) **Success Rate** ($\text{SR}_{\text{train-env}}$) is the fraction of evaluation episodes satisfying the above success conditions in the restored training environment,

including native noise. 2) **Cumulative Reward** is the task-specific evaluation return and is never pooled across tasks. 3) **Peak-Efficiency Steps** is $T_{\max} = \arg\max_{t \in \mathcal{T}} \mathbb{E}[R_t]$, the evaluated training step attaining the highest mean return. All-task and family success rates are task-wise arithmetic means. 4) **Perturbation Robustness** evaluates separate action, observation, and muscle-dynamics sweeps using Gaussian action/observation noise and uniformly sampled multiplicative muscle-force-gain perturbations. Action/dynamics scales are (0, 0.05, 0.10, 0.15, 0.20) and observation scales are (0, 0.02, 0.05, 0.08, 0.10); the score is the normalized trapezoidal area under success versus perturbation scale, averaged equally over tasks and perturbation types. 5) **Activation Cost**, $E = (TM)^{-1} \sum_{t,m} a_{m,t}^2$, is the mean squared muscle activation and serves as an effort proxy. 6) **Joint Smoothness** uses $j_{q,t} = (v_{q,t} - 2v_{q,t-1} + v_{q,t-2})/\Delta t^2$; its squared magnitude is averaged over joints and time before logarithmic display. Both are interpreted only for successful, behaviorally comparable policies. 7) **EMG-Envelope Similarity**, following the activation-envelope interpretation used in MuscleMimic [15], computes per-muscle Pearson correlation $r_m = \text{corr}(a_{\text{sim},m}, e_{\text{EMG},m})$ and reports $r_{\text{EMG}} = M^{-1} \sum_m r_m$. Signals are resampled to 101 gait-cycle points, min-max normalized per muscle, cycle-averaged, and cyclically shifted to maximize correlation, so the metric reflects phase-optimized waveform shape rather than absolute amplitude or timing.

Baselines. We consider **five** control paradigms within a common policy-action-objective formulation. Let p index the paradigm, s_t denote the simulator state, o_t its observation, and c_t optional reference or prior information:

$$\begin{aligned} \mathbf{u}_t &\sim \pi_{\theta}^p(\cdot | o_t, c_t), \\ \mathbf{a}_t &= F_p(o_t, \mathbf{u}_t, c_t) \in [-1, 1]^M, \\ J_p(\theta; \mathbf{w}_{p,k}) &= \mathbb{E}_{\tau \sim P, \pi_{\theta}^p, F_p} \left[\sum_{t=0}^{T-1} \gamma \tilde{r}_{p,t} \right], \\ \tilde{r}_{p,t} &= \mathbf{w}_{p,k}^\top \mathbf{f}_{p,t} - \lambda_{p,k} \Omega_{p,t} + \beta_p \mathcal{H}_{p,t}. \end{aligned} \quad (1)$$

Here P is the shared transition model, M the number of muscles, $\mathbf{u}_t$ the learned command, and $\mathbf{a}_t$ the executed excitation. The action map F_p , reward features $\mathbf{f}_{p,t}$, penalties $\Omega_{p,t}$, and entropy $\mathcal{H}_{p,t}$ specify each paradigm; adaptive weights remain fixed within stage k .

1) **Reward-based RL** uses fixed task rewards with direct muscle actions or an algorithm-specific action map F_p . We evaluate PPO [22], SAC [23], DepRL [24], and DynSynSAC [25], covering standard, exploration-enhanced, and action-structured methods. 2) **Agentic reward tuning** uses DepRL as the base learner and updates reward coefficients through bounded reasoner feedback, $\mathbf{w}_{k+1} = U(\mathbf{w}_k, \mathbf{h}_k)$. We evaluate DeepSeek-V4-Flash [30] and GPT-6 Astra [31] as reasoners. 3) **Latent-action RL** filters commands through an expert-trained, state-conditioned encoder-decoder. The latent dimension constrains the representation bottleneck while the executed muscle-action space remains unchanged. 4) **Imitation-prior control** uses a pretrained MuscleMimic [15] policy, introducing reference information through c_t and motion-tracking objectives. 5) **Residual adaptation over**

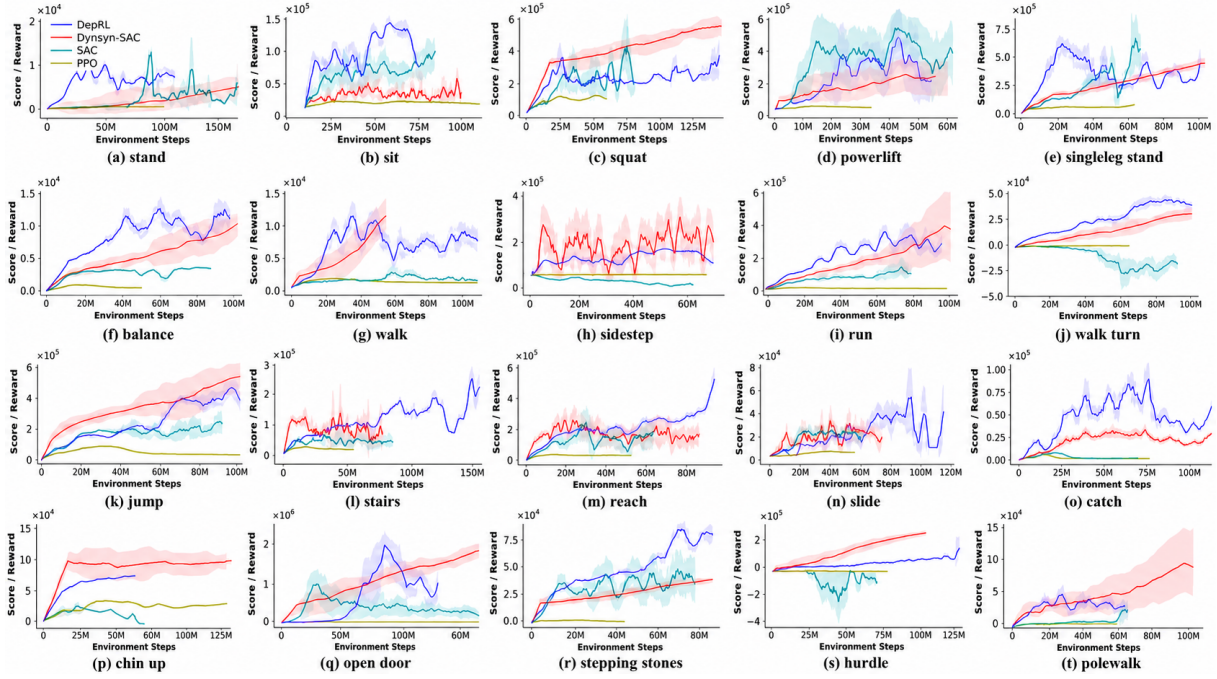


Fig. 2. Task-specific reward-learning curves for 20 of 22 tasks. Four reward-based algorithms show task-dependent advantages. Shading shows across-run variation; Table II separately reports success, nonzero-success coverage, and robustness over all 22 tasks.

TABLE II

FULL-SUITE POINT ESTIMATES. SR RETAINS TRAINING NOISE. MEANS WEIGHT TASKS EQUALLY. ROBUSTNESS AREAS ARE NORMALIZED TO 0–100; AVG. WEIGHTS THE THREE PERTURBATION TYPES EQUALLY.

COVERAGE: SR > 0. BOLD: ROW BEST.					
Group	Metric	DynSyn-SAC	DepRL	SAC	PPO
SR (%) ↑	Stabilization (6)	57.33	50.33	42.33	0.00
	Locomotion (6)	81.33	41.33	0.00	0.00
	Interaction (10)	31.20	51.40	2.80	7.40
	Average (22)	52.00	48.36	12.82	3.36
Robust. ↑	Action	46.66	53.06	15.36	3.10
	Observation	43.37	38.10	4.26	3.40
	Dynamics	34.75	39.84	14.50	2.92
	Average	41.59	43.67	11.37	3.14
Coverage ↑ SR > 0		16/22	22/22	4/22	1/22

imitation priors combines a frozen prior with a bounded, task-specific muscle-action correction:

$$\mathbf{a}_t^{\text{full}} = \text{clip}\left(\mathbf{a}_t^{\text{base}} + \alpha \text{clip}(\mathbf{a}_t^{\text{res}}, -1, 1), -1, 1\right). \quad (2)$$

The four **reward-based** algorithms are trained and evaluated on all 22 tasks to form the full-suite leaderboard. **Agentic** reward tuning and **latent-action RL** are focused studies; **MuscleMimic** is evaluated on five tasks, and **residual adaptation** on walk, run, and stairs. These studies differ in task coverage, and some modify reward coefficients or introduce reference/prior objectives, so they are reported separately from the full-suite ranking.

Results & Analysis. We use MSK-Bench to address three questions. First, *do aggregate scores adequately characterize controller performance?* Task-family comparisons reveal ranking reversals and differences in coverage and robustness, while focused studies examine the effects of reward adaptation and action compression (**T1**). Second, *where do motion*

priors succeed, and how can their failures be characterized? Task-specific robustness tests reveal contrasting outcomes across movements, and tracking-error analysis localizes divergence on stairs without establishing its cause (**T2**). Third, *What aspects of muscle use are omitted by task-level scores, and how can they be assessed?* Paired success and EMG results expose differences beyond task completion, while muscle-resolved, task-dependent, and anatomical analyses characterize where those differences occur (**T3**).

T1–Breadth: controller rankings depend on task family and metric. Fig. 2 shows task-dependent reward-learning trajectories for the four reward-based algorithms on 20 of the 22 tasks. Over the displayed overlapping training ranges, DynSyn-SAC generally attains higher returns than DepRL on jump and hurdle, whereas DepRL attains higher returns on walk-turn and catch (Fig. 2j,k,o,s). SAC also remains competitive on powerlift (Fig. 2d). Thus, the learning curves reveal differences not only across task families, but also among tasks within the same family. Tab. II separately quantifies success, coverage, and robustness over the full suite. DynSyn-SAC attains higher mean success, driven primarily by locomotion, whereas DepRL performs better on interaction tasks. DepRL achieves nonzero observed success on 22/22 tasks, compared with 16/22 for DynSyn-SAC, and has higher action- and dynamics-robustness scores; DynSyn-SAC has a higher observation-robustness score. These results concern separately trained task-specific policies and show why mean success alone is insufficient to characterize controller performance.

Complementary diagnostics: objectives and action representations. The following focused studies examine additional design choices. In the original Agentic-DepRL setting, the reasoner receives mean forward speed and torso pitch

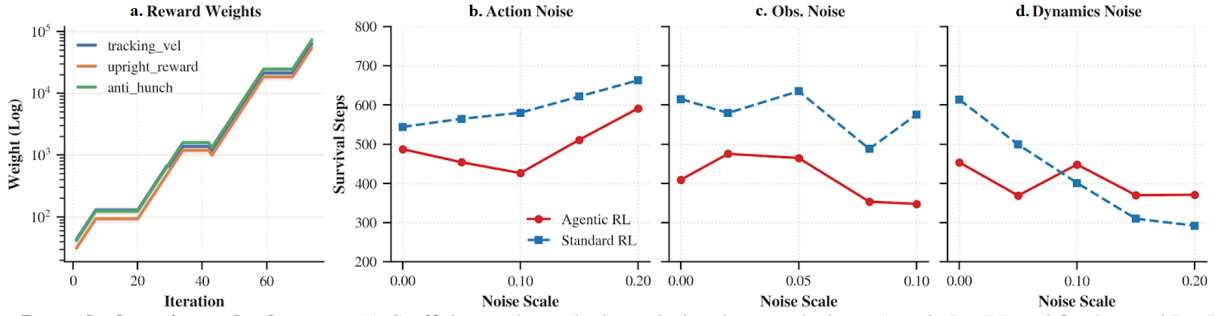


Fig. 3. **Reward adaptation and robustness.** (a) Coefficient updates; (b–d) survival under perturbations. Agentic-DepRL and fixed-reward DepRL share learner, architecture, and training budget. Each point averages 50 held-out seeds; survival is capped at 1000 steps.

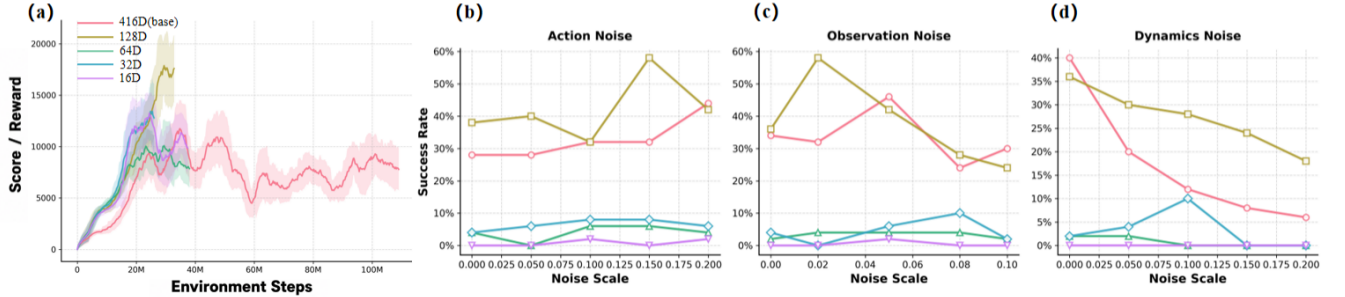


Fig. 4. **Action compression.** Learning and perturbation tests for 416 actions and 128-, 64-, 32-, and 16-dimensional representations. The focused study improves at 128 dimensions but degrades under stronger compression; it is not a full-suite comparison.

TABLE III
MATCHED REASONER COMPARISON.

Setting	Survival (steps) $\uparrow$	Tracking RMSE $\downarrow$	Activation $\downarrow$
Fixed reward	417	0.4446	0.4769
DeepSeek	246.6	0.4355	0.3999
GPT-6 Astra	360	0.4324	0.3879

and updates reward coefficients every 20,000 steps, with each nonzero-weight change limited to 20%. Despite these bounded updates, the unnormalized coefficient magnitudes grow over time (Fig. 3a). Fixed-reward DepRL maintains higher survival under the tested action and observation perturbations, whereas agentic tuning yields higher survival at the larger dynamics-perturbation scales (Fig. 3b–d). Reward adaptation therefore provides no uniform survival advantage.

In a matched comparison, DeepSeek and GPT-6 Astra use the same DepRL learner, architecture, training budget, and evaluation, with an identical reward-coefficient constraint $\|w\|_1 = 40$. GPT-6 Astra has better reported point estimates than DeepSeek for survival, tracking RMSE, and activation (Tab. III). Relative to fixed-reward DepRL, however, it yields shorter survival (360 vs. 417 steps) despite lower RMSE and activation, illustrating a trade-off among the reported metrics.

A separate diagnostic tests GPT-6 Astra without training or a learned low-level controller. It directly outputs 354-dimensional muscle excitations at 5 Hz in a 100-Hz simulation, with each action held for 0.2 s. One seed-0 rollout is evaluated for each of walk, run, and stairs; all three terminate in falls within the 3-s horizon (Tab. IV). These outcomes characterize the tested direct-action configuration rather than establishing a general limitation of GPT-6 control.

Finally, latent-action RL uses an expert-trained, state-conditioned encoder–decoder while retaining the executed muscle-action space. The 128-dimensional setting achieves

TABLE IV
GPT-6 DIRECT MUSCLE-CONTROL DIAGNOSTIC. DIV.: FIRST SUSTAINED TRACKING DIVERGENCE; N/A INDICATES THAT THE FIXED DIVERGENCE CRITERION WAS NOT MET.

Task	Duration (s) $\uparrow$	Div. (s) $\uparrow$	Tracking RMSE $\downarrow$	Outcome
Walk	0.75	0.53	0.158	Fall
Run	2.02	0.94	0.653	Fall
Stairs	0.80	N/A	0.173	Fall

higher returns over the displayed overlapping training interval and higher success at several perturbation scales than the 416-dimensional baseline, whereas stronger compression degrades performance (Fig. 4). The reported reconstruction and explained-variance statistics also deteriorate under stronger compression, consistent with a representation trade-off rather than an unrestricted benefit from dimensionality reduction. Together, these focused studies reinforce the need to evaluate reward adaptation and action representations against task-specific outcomes and multiple metrics.

T2-Diagnosis: task-specific tests characterize a stair-control failure. Under the tested dynamics perturbations, MuscleMimic generally achieves higher measured success than DepRL on stand, jump, walk, and run (Fig. 5a–d), with a smaller gap on jump. On stairs, however, MuscleMimic has no observed successes at any tested scale, including zero dynamics perturbation, whereas DepRL retains nonzero success (Fig. 5e). The stair limitation is therefore present at the zero-perturbation setting, rather than arising only when perturbations increase. Success on the other evaluated skills does not establish suitability for this task.

The reported stair-tracking errors characterize increasing deviation from the reference (Fig. 5f). Between steps 150 and 300, absolute-site error rises from 0.35 to 1.31, and root-XYZ error from 0.25 to 0.77. A mismatch between reference

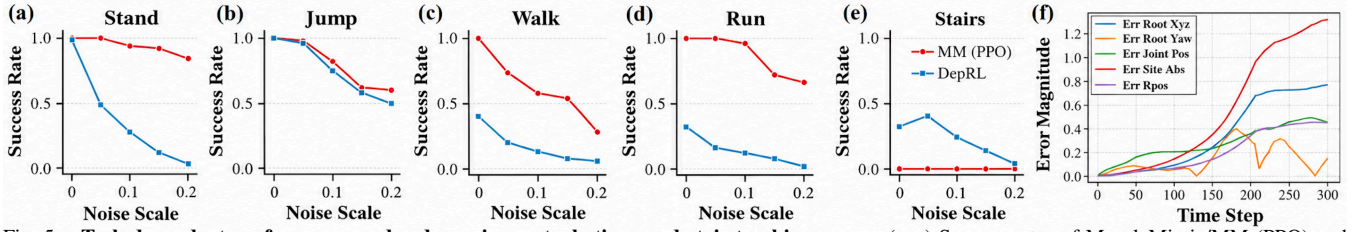


Fig. 5. **Task-dependent performance under dynamics perturbations and stair-tracking errors.** (a–e) Success rates of MuscleMimic/MM (PPO) and DepRL. MuscleMimic has no observed stair successes at any tested scale, including zero dynamics perturbation. (f) Reported tracking-error components on stairs characterize deviation from the reference, without identifying a timing mismatch or the cause of task failure.

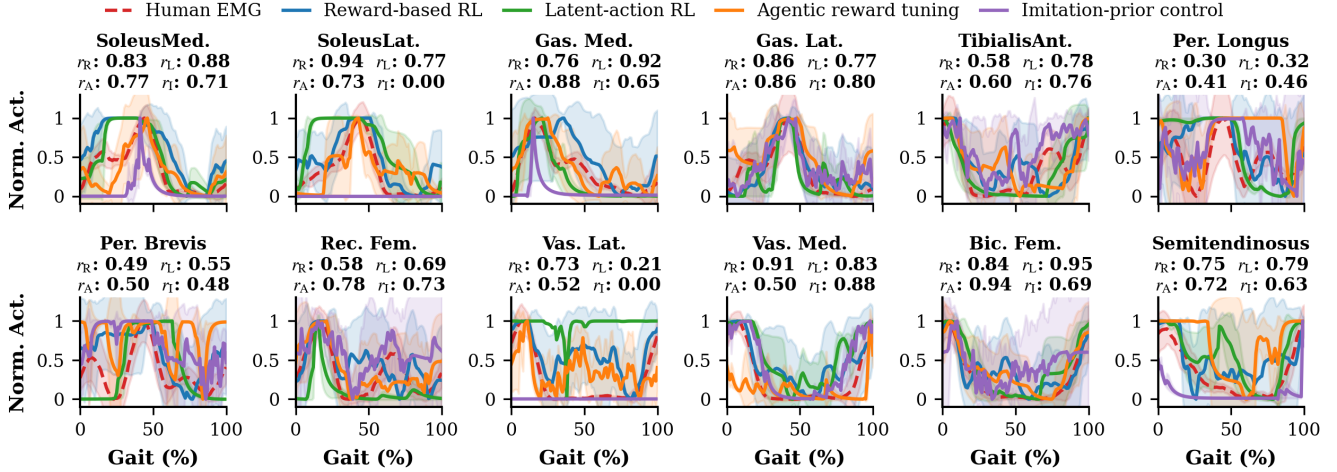


Fig. 6. **Walking EMG-envelope comparison.** Human recordings [32] versus simulated activations. Normalized, phase-optimized correlations (r_R , r_L , r_A , r_I) correspond to reward-based RL, latent-action RL, agentic tuning, and imitation-prior control.

progression and executed motion is one possible explanation, but these errors neither establish a timing mismatch nor identify the cause of task failure. Motivated by this task-specific failure, Sec. IV evaluates a combined adaptation pipeline involving residual actions, task-objective changes, and reference pacing; **T3** assesses task success and EMG-envelope similarity separately.

T3-Physiology-aware evaluation: task success does not characterize muscle use. MSK-Bench complements task outcomes with muscle-resolved diagnostics and, where human recordings are available, EMG-envelope comparisons. These analyses distinguish task completion from the muscle activity underlying the reported behavior.

Task outcomes and human-reference agreement. Tab. V illustrates this distinction using the combined adaptation pipeline described in Sec. IV. Walking retains 100% SR while correlations for the three selected muscles increase from 0.71/0.69/0.63 to 0.75/0.80/0.67, revealing changes not captured by success alone. Running improves from 90% to 100% SR and from 0.43 to 0.77 in mean EMG correlation. In contrast, stairs recovers from 0% to 100% SR while mean correlation changes only from 0.51 to 0.52, remaining below DepRL’s 0.56. Thus, an unchanged success rate can accompany changes in muscle-level agreement, while a large success gain can accompany only a small numerical change in the EMG metric.

Muscle-specific differences across control settings. Fig. 6 further resolves walking agreement by muscle. Latent-action RL has higher correlation than reward-based RL for biceps femoris (0.95 vs. 0.84), but lower correlation for vastus

lateralis (0.21 vs. 0.73). Such reversals reveal muscle-specific differences that an aggregate score does not describe. Human references are Gait120 [32] for walking and stairs and the 3-m/s recordings of Van Hooren and Meije [33] for running. Because the signals are normalized and phase-optimized, these comparisons assess activation-envelope shape rather than absolute amplitude or timing.

Task and anatomical context. Complementary diagnostics characterize movement fluctuations and recruitment across task settings. Fig. 7 reports task-dependent joint-jerk levels with overlapping ranges across task families, providing movement-level context rather than a cross-task ranking of control quality. In powerlift, muscle-family recruitment varies across loads: trunk/back activation mass is 30.1, 50.4, and 46.0 at 0.05, 5, and 10 kg, respectively (Fig. 8). These profiles describe load-specific muscle use over the recorded motions, rather than a uniform increase in activation or agreement with human recordings. Fig. 9 extends this analysis to anatomical resolution. The 700-muscle model reports additional neck, deep-trunk, and deep-scapular recruitment groups, showing that the available anatomical representation also determines what muscle-use diagnostics can reveal. These model-specific profiles do not establish greater human similarity. Together, the analyses extend evaluation beyond task completion to muscle-wise human-reference agreement and task- and anatomy-specific descriptions of muscle use.

The remaining diagnostics localize differences after task success is established. In powerlift, the axial/core-to-upper-extremity activation ratio increases from 1.42 to 2.27 with load (Fig. 8). Walking correlations vary by muscle, with

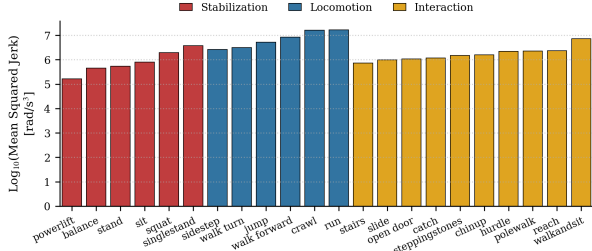


Fig. 7. **Task-wise joint smoothness.** Bars report $\log_{10}$ mean squared angular jerk. Stabilization shows the lowest measured jerk, locomotion the highest, and interaction intermediate values. These characterize motion fluctuations, not success or human similarity.

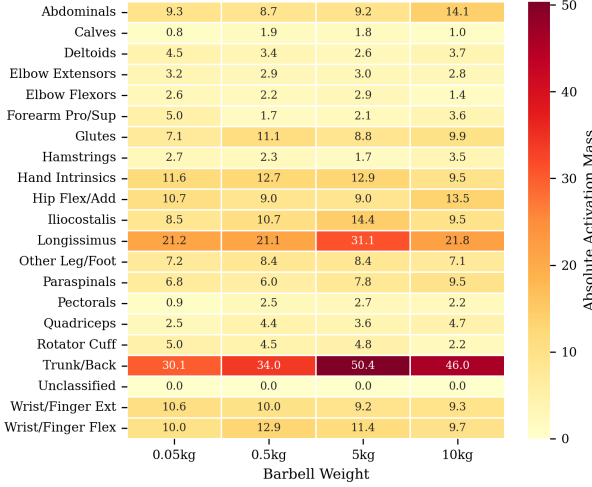


Fig. 8. **Load-dependent recruitment in powerlift.** Activation mass accumulated by muscle family under different barbell weights. Totals describe muscle use over the recorded motion, not instantaneous activation or agreement with human recordings.

reward-based RL higher for lateral soleus (0.94 vs. 0.00) but the imitation prior higher for tibialis anterior (0.76 vs. 0.58). The 700-muscle model additionally recruits neck, deep-trunk, and scapular groups, while joint jerk characterizes motion fluctuations (Figs. 9 and 7).

IV. RESIDUAL ADAPTATION CASE STUDY

T2 identifies stairs as an adaptation target; T3 asks separately whether adaptation recovers the task and improves EMG agreement. We use residual adaptation to demonstrate this diagnosis-to-validation transition, not as a standalone state-of-the-art method. Fig. 10 shows the controller and example motions. Specializing Eq. (1), a frozen MuscleMimic policy [15] supplies base actions, and a trainable 3-layer MLP supplies the bounded corrections defined in Eq. (2).

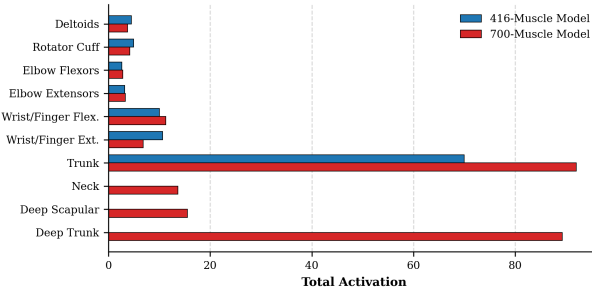


Fig. 9. **Anatomical resolution and powerlift recruitment.** Activation mass for the 416- and 700-muscle models. The latter adds neck, deep-trunk, and deep-scapular groups, expanding available recruitment pathways; this does not establish greater human similarity.

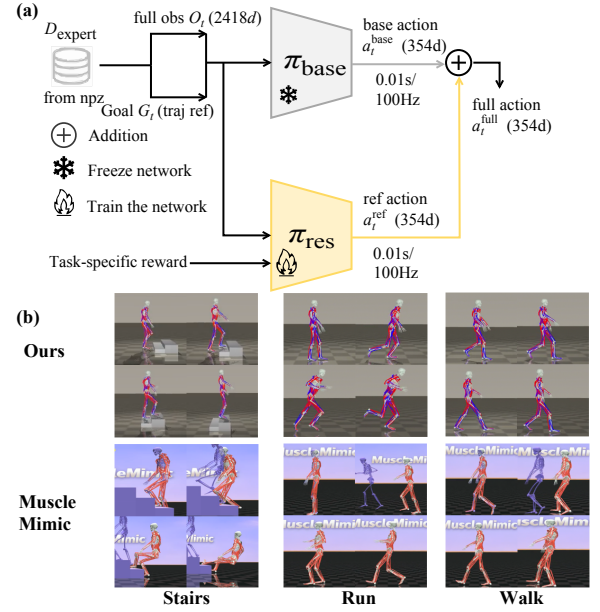


Fig. 10. **Residual architecture and motions.** Frozen prior plus trainable correction; blue skeletons are references. The original lower-branch label “ref action” denotes the residual correction in our equation. The 354d labels exclude hand actuation. Rewards and reference timing also change.

TABLE V

RESIDUAL ADAPTATION RESULTS. MM: MUSCLEMIMIC. SR RETAINS TRAINING NOISE. EMG r REPORTS SOLEUSMED./BIC.FEM./SEMITEND. FOR WALKING AND 12-MUSCLE PHASE-ALIGNED MEANS FOR RUNNING AND STAIRS. BOLD: BEST IN EACH COMPARISON.

Task	SR (%) $\uparrow$		EMG r $\uparrow$		
	MM	Residual	MM	DepRL	Residual
Walking	100	100	0.71/0.69/0.63	–	0.75/0.80/0.67
Running	90	100	0.43	0.70	0.77
Stairs	0	100	0.51	0.56	0.52

Besides the residual correction, adaptation changes task objectives and timing: stairs relaxes vertical tracking and adapts reference pacing; walking uses phase-dependent residual authority and activation regularization; running adjusts flight-phase rewards and penalizes ground-reaction forces. A small L2 residual penalty ($c_{\text{res}} = -0.02$) replaces the generic action-rate and activation-cost terms. Results therefore characterize the combined pipeline.

Task-specific adaptation. For stairs, planar root error controls reference progression: below 0.4 m the reference advances normally; otherwise it slows by 50%. Vertical root displacement is excluded from tracking, and progress targets the reference state 1 s ahead. Walking uses residual authority 0.6 in swing and 0.2 in stance and penalizes activation magnitude and temporal change. Running halves flight-phase tracking penalties, rewards toe-off, and softly penalizes vertical GRF above $2.5 \times$ body weight. These choices permit larger swing corrections, preserve support, and avoid treating the residual study as an unchanged-objective ablation.

Tab. V summarizes task recovery and EMG agreement. Walking retains 100% SR while improving correlations for the three selected muscles, and running improves both SR and mean EMG correlation. Stairs shows the largest success gain but only a 0.01 correlation increase over MuscleMimic,

remaining below DepRL (0.56). Thus, task recovery does not imply uniformly better human EMG agreement. Stair height rises from 0.94 to 1.21 m and tracking error falls from 0.1968 to 0.1079 rad; running tracking error falls from 0.2267 to 0.1338 rad and peak GRF from 6.40 to 6.07 times body weight. Together, these results close the T1–T3 chain: broad evaluation motivates task-specific diagnosis, which in turn motivates multi-metric validation after adaptation.

V. CONCLUSION

In this work, we introduced **MSK-Bench**, a full-body musculoskeletal benchmark built on a 416-muscle humanoid, with 22 tasks spanning stabilization, locomotion, and contact-rich interaction. MSK-Bench provides a unified evaluation protocol covering task performance, robustness, activation cost, joint smoothness, and EMG-envelope similarity. Our results show that embodiment-aware exploration and action-structured representations improve task coverage in the evaluated settings; motion-prior control performs strongly on several reference-compatible skills but fails on the tested stair task; and physiology-aware diagnostics reveal muscle-level differences not captured by task success alone. We further use residual adaptation as a case study that combines a motion prior with task-specific corrections to recover task performance while separately assessing EMG agreement. MSK-Bench provides a testbed for physiologically grounded whole-body control.

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